

Chapter 22

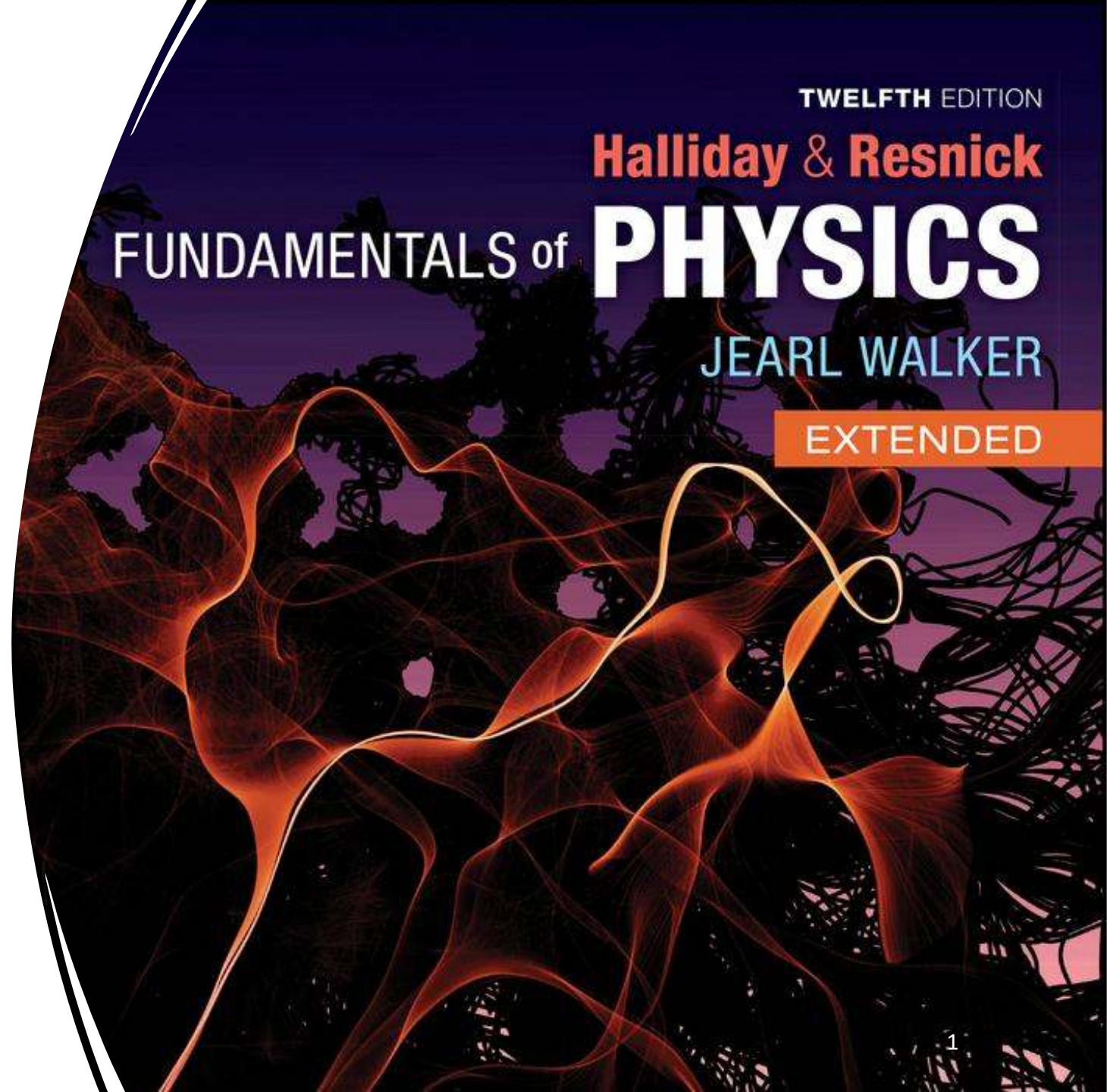
Electric Fields

22.1 The electric field

22.2 The electric field due to a charged particle

22.6 A point charge in an electric field

Fundamentals of Physics, Twelfth Edition.
Halliday & Resnick, Walker



Section 22.1 The Electric Field

Electrostatic Forces at a Distance

charge disturb place
around, and feel presence
of e field



How does particle 1 “know” of the presence of particle 2? That is, since the particles do not touch, how can particle 2 push on particle 1—how can there be such an action at a distance?

Will be asked about it again

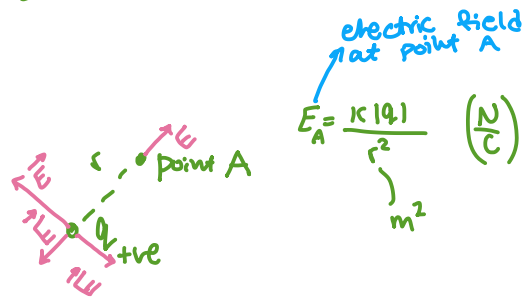
Field: disturbance

PHY105: gravitational field

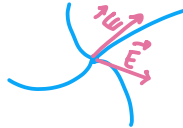
↳ for every mass
disturb space around it → creating field

presence of charge creates an electric field
magnet creates magnetic field

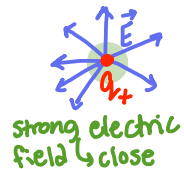
charge: q
 $W = mg$
minimum 2 things (pairs)



Electric field lines never intersect

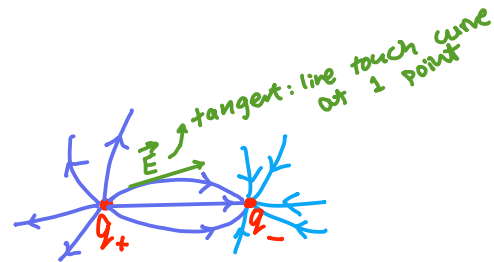


E field start from charge to ∞



also lines close so strong

E field from ∞ to charge



Doesn't disturb itself

Wed 7 Jan

$$F = mg$$

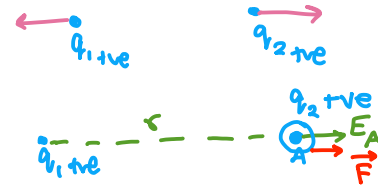


asked about it 100%

$$\vec{F} = q\vec{E}$$

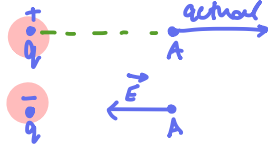
q_{+ve}
 $\vec{F} \propto \vec{E}$
Same direction

q_{-ve}
 $\vec{F} \propto \vec{E}$
Opposite direction



Force on $q_2 = q_2 \times E_A$
 $= \frac{q_2 K |q_1|}{r^2}$
 $= \frac{K |q_1 q_2|}{r^2}$

E field due to a point charge



mistake is to draw on q_1
rather must be on point
since charge doesn't disturb itself

uniform electric field
field lines: parallel & = magnitude



Electrostatic Forces at a Distance



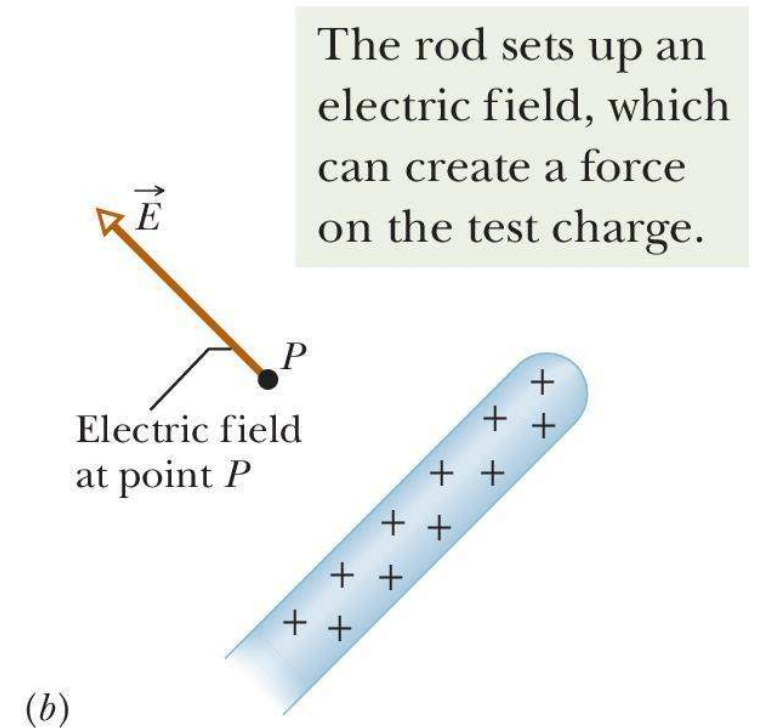
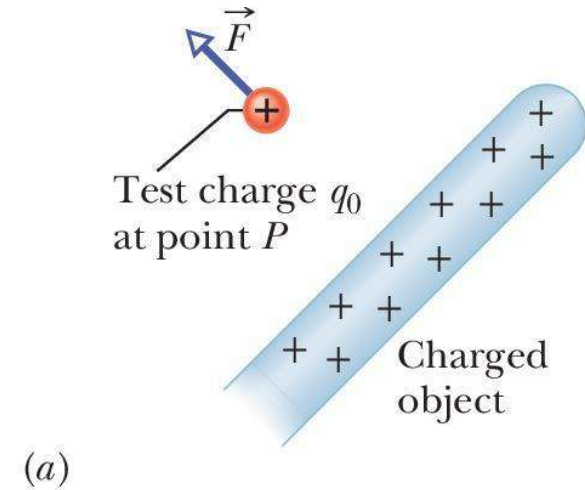
- The explanation is this: Particle 2 sets up an **electric field** at all points in the surrounding space, even if the space is a vacuum
- If we place particle 1 at any point in that space, particle 1 knows of the presence of particle 2 because it is affected by the electric field particle 2 has already set up at that point.
- Thus, particle 2 pushes on particle 1, but not by touching it as you would push on a coffee mug. Instead, particle 2 pushes by means of the electric field it has set up.

Electric Field Vector Expression

The **electric field** $\vec{E}$ at any point is **defined** in terms of the **electrostatic force** $\vec{F}$ that would be exerted on a **positive test charge** q_0 placed there:

$$\vec{E} = \frac{\vec{F}}{q_0}$$

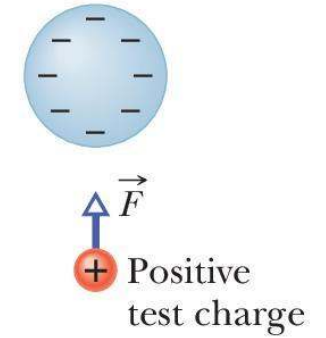
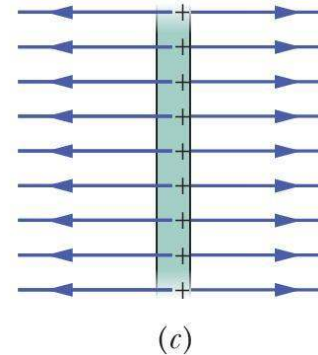
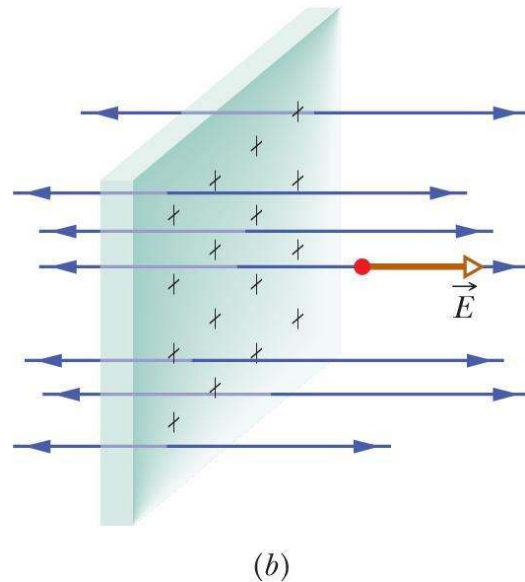
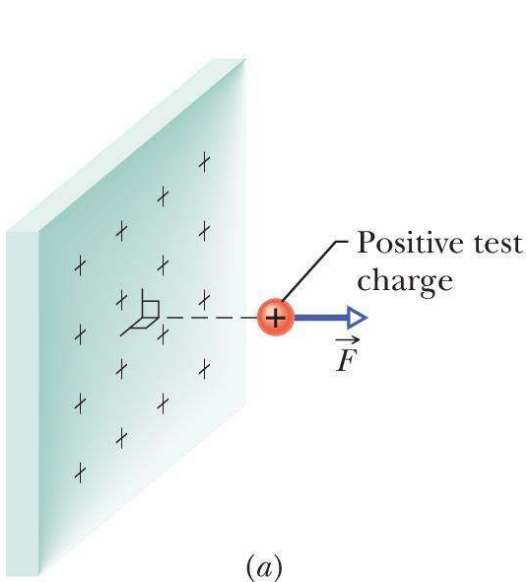
SI unit: N/C



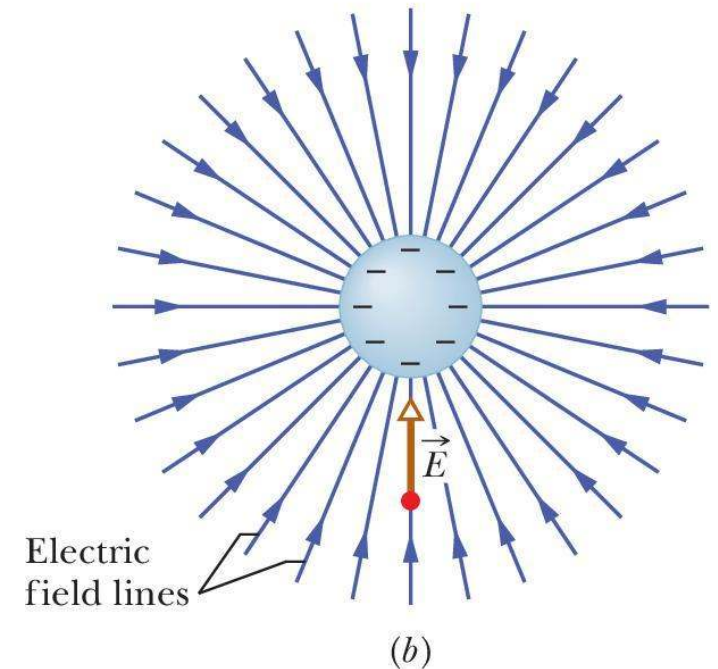
Electric Field Lines

Electric field lines help us visualize the direction and magnitude of electric fields.

1. The electric field vector at any point is **tangent to the field line** through that point and in the same direction.

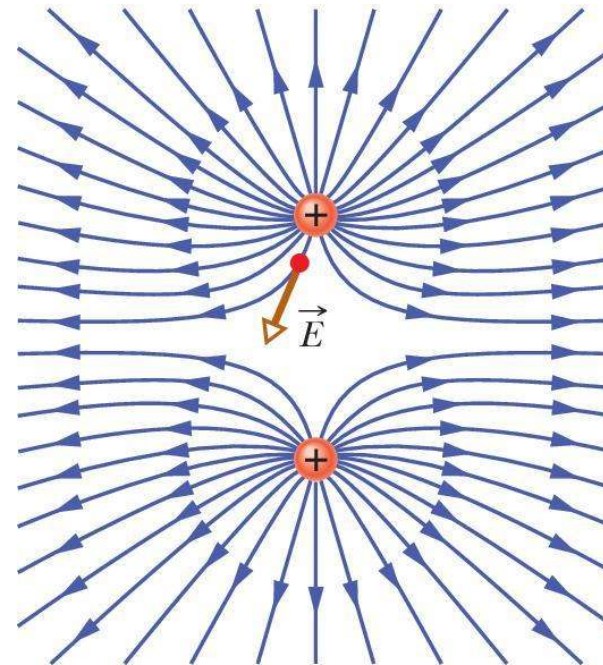
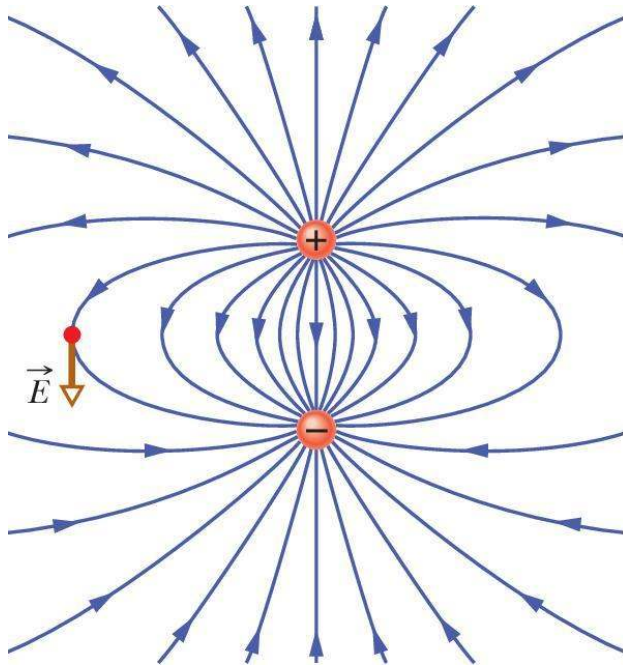


(a)

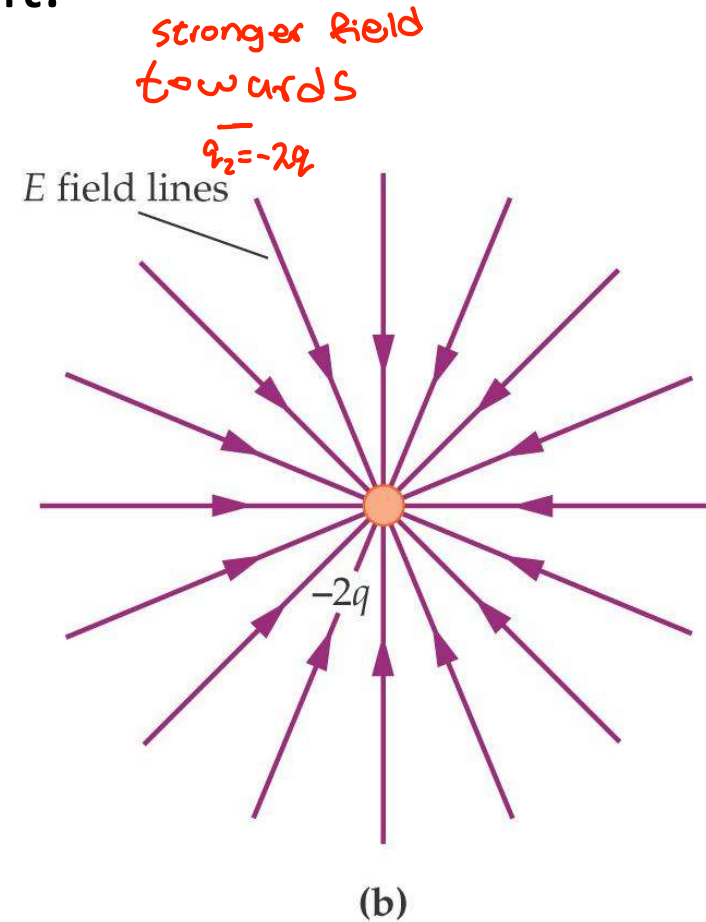
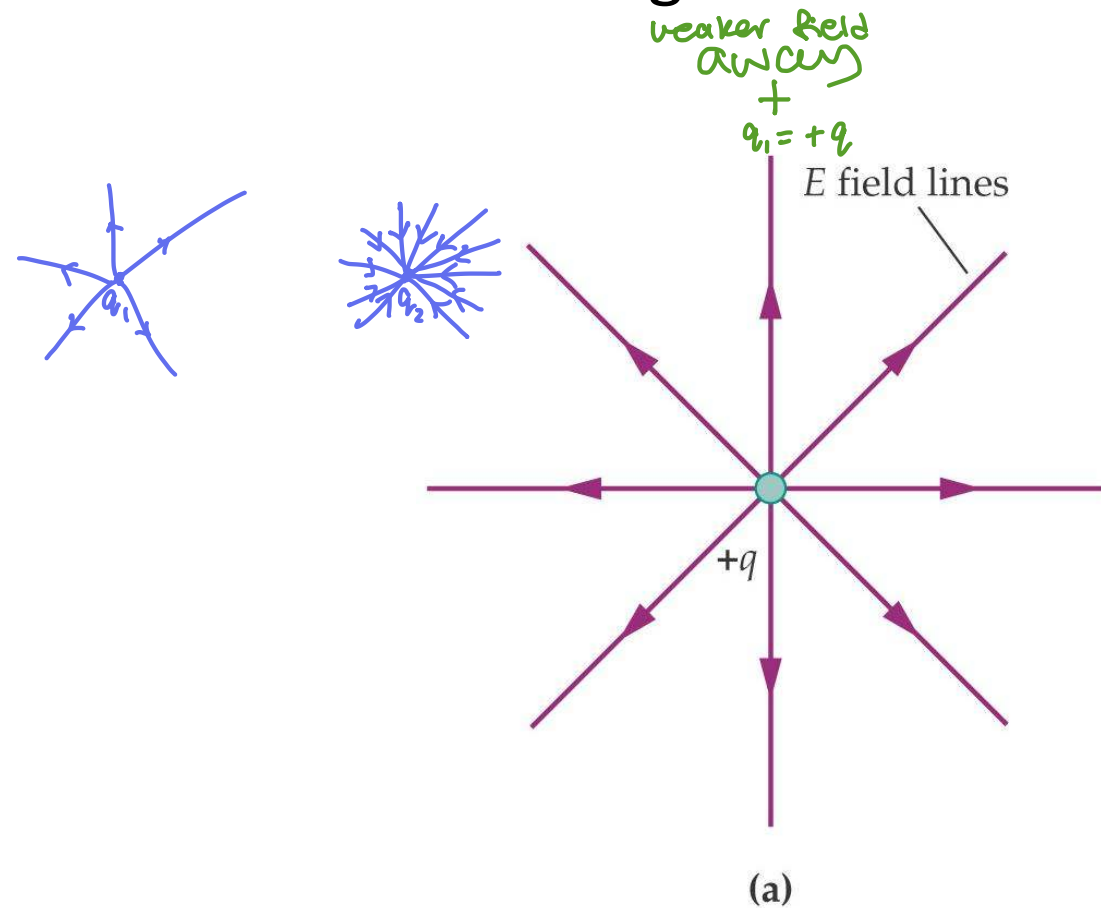


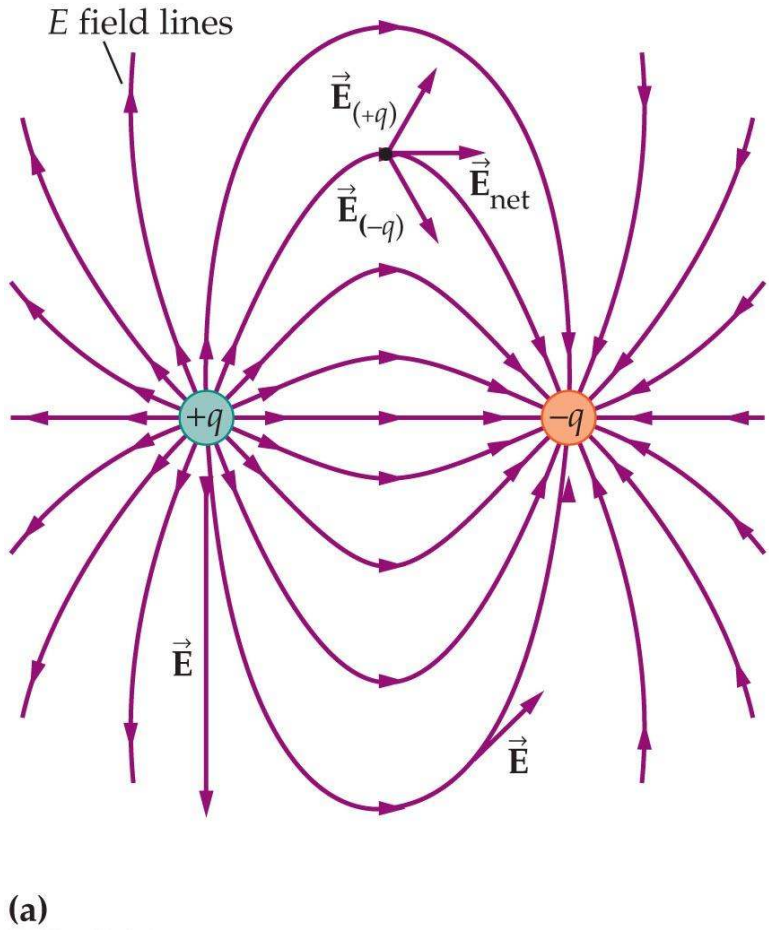
(b)

2. The **density of field lines** in that region is **proportional to the magnitude** of the electric field there. A closer spacing means a larger field magnitude
3. Electric field lines **extend away from positive charge (where they originate)** and **toward negative charge (where they terminate)**.

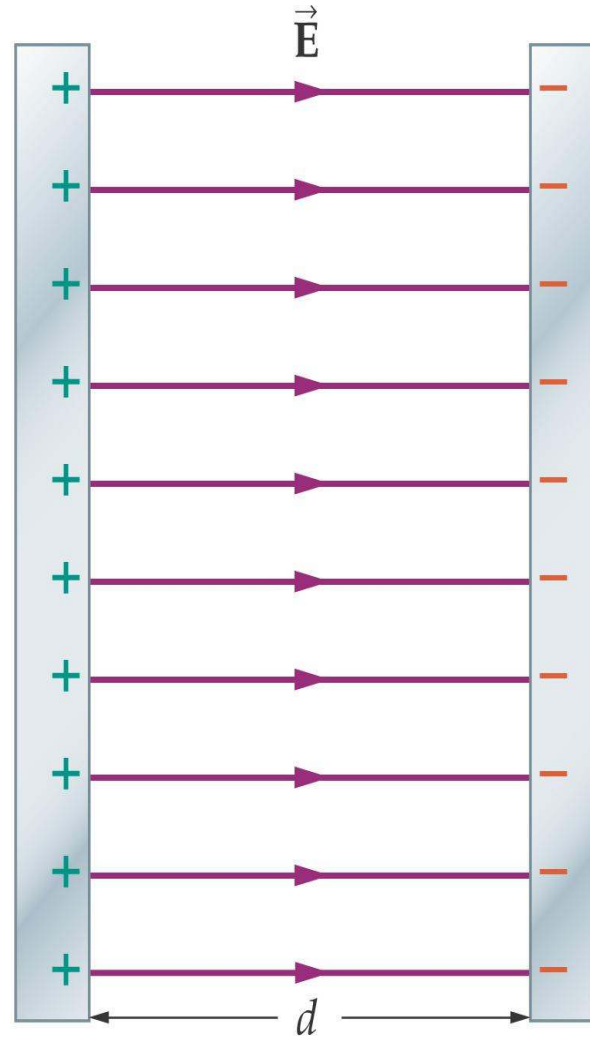


The charge on the **right** has **twice the magnitude** of the charge on the left (and opposite in sign), so there are **twice as many field lines**, and they point towards the charge rather than away from it.





- The electric field around a **dipole**



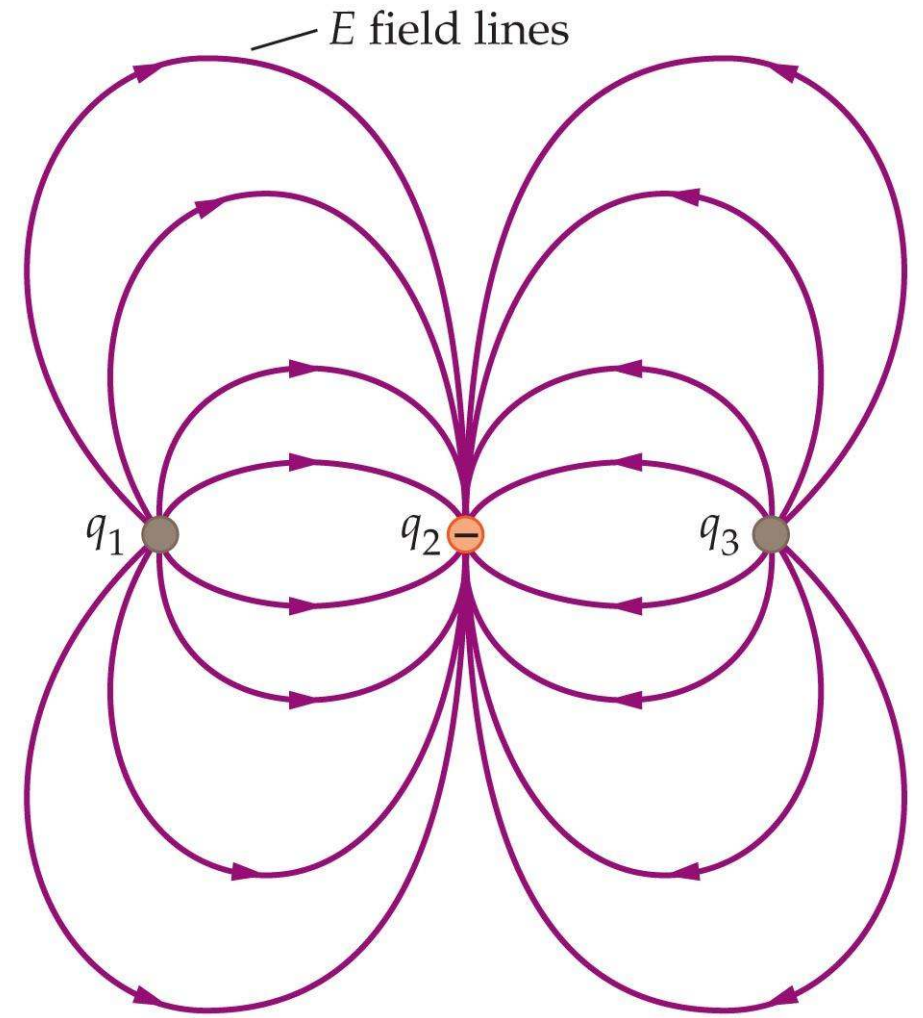
- A **uniform Electric field**: same magnitude and direction at every point within the field

Exercise: The electric field lines surrounding three charges are shown in the figure. The charge at the center is $q_2 = -10.0 \mu\text{C}$

- a) What is the sign of q_1 and q_3 ?
- b) Find q_1
- c) Find q_3

8 lines away from q_1 & q_3 so can label it as q
but q_2 is opp sign (-) & double number so $-2q$

$$\begin{aligned} q_1 &= +q & +5\mu\text{C} \\ q_2 &= -2q & = -10\mu\text{C} \\ q_3 &= +q & +5\mu\text{C} \end{aligned}$$



Solution: (a) The electric field lines begin at q_1 and q_3 and end at q_2 . The rules of drawing electric field lines indicate that the charges q_1 and q_3 must be positive.

(b) The charge q_1 has 8 lines leaving it but q_2 has 16 lines entering it. Because 8 is half of 16, and since the number of lines entering or leaving a charge is proportional to the magnitude of the charge, the magnitude of q_1 is one-half of q_2 $(+5 \mu\text{C})$

(c) By the same reasoning of part **(b)**, the magnitude of q_3 is $(+5 \mu\text{C})$

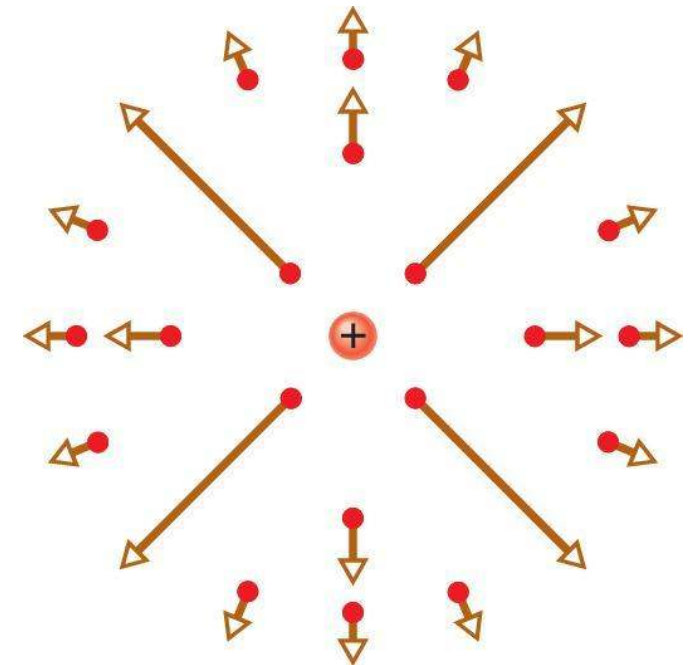
Note: When drawn correctly, an electric field diagram can reveal a wealth of information in a concise approach.

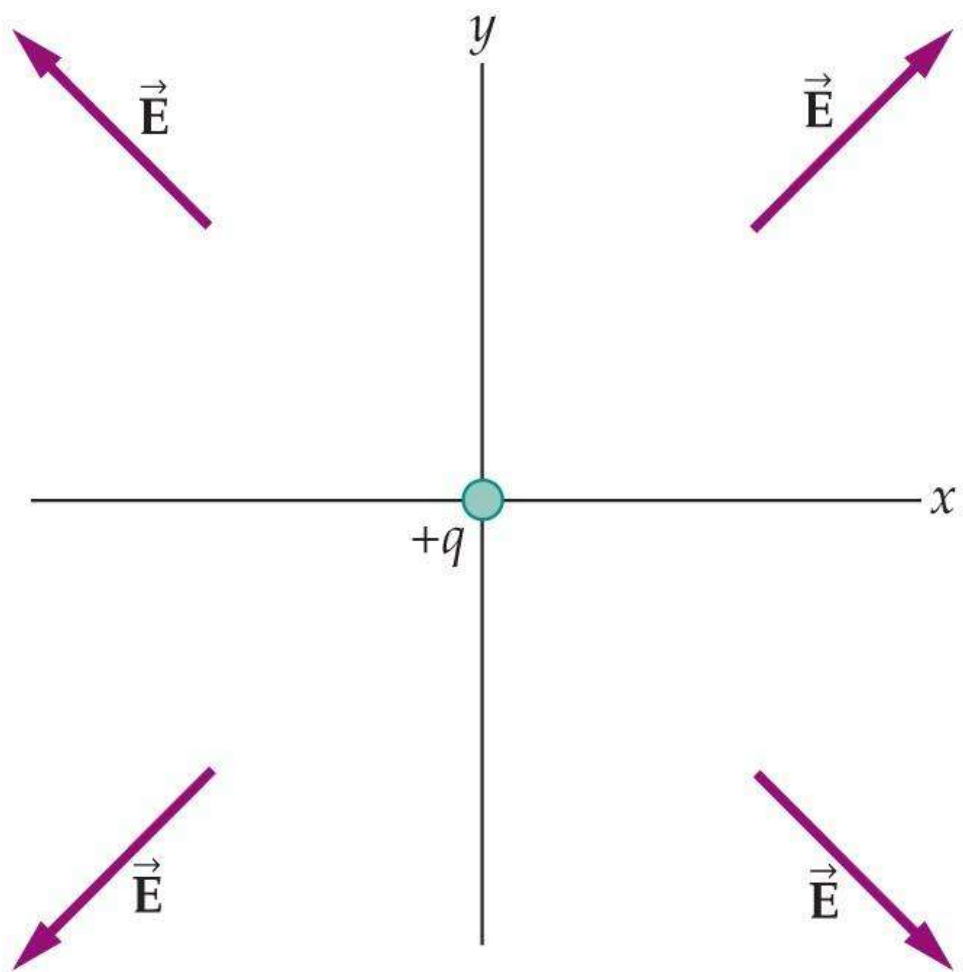
Section 22.2 The Electric Field Due to a Charged Particle

The **magnitude** of the electric field $\vec{E}$ set up by a particle with **charge** q at **distance** r from the particle is

$$E = \frac{1}{4\pi\epsilon_0} \frac{|q|}{r^2}$$

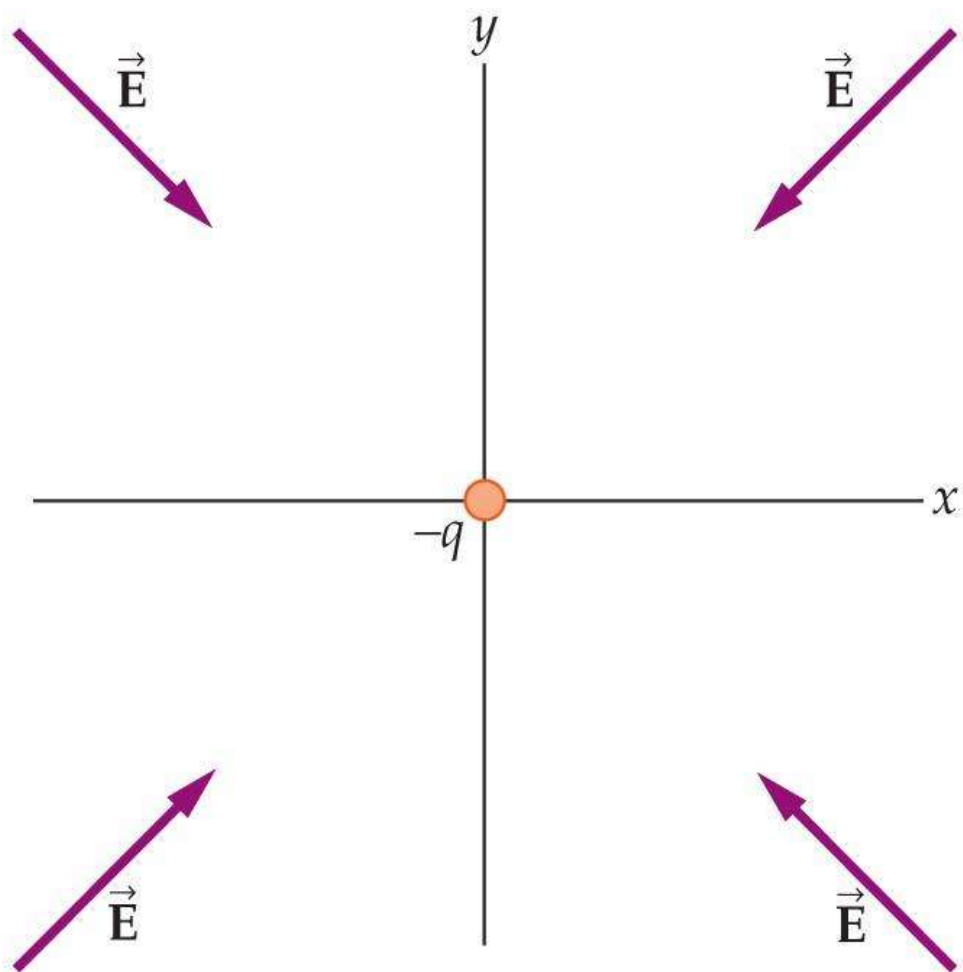
- The **electric field vectors** set up by a **positively charged** particle all point **directly away from the particle**.
- Those set up by a **negatively charged** particle all point directly **toward** the particle.





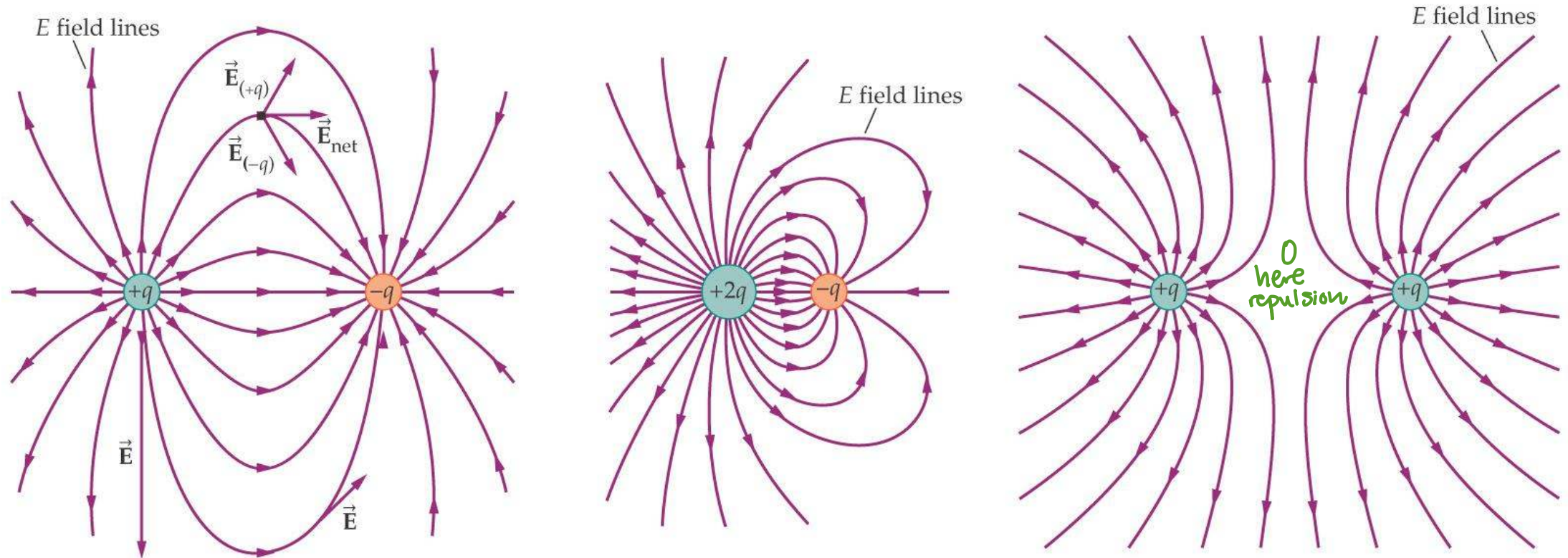
(a)

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(b)

If **more than one** charged particle sets up an electric field at a point, the **net electric field** is the **vector sum** of the individual electric fields—**electric fields obey the superposition principle**.



There is only one **point** within the figures above where the **field is zero**. Can you find it?

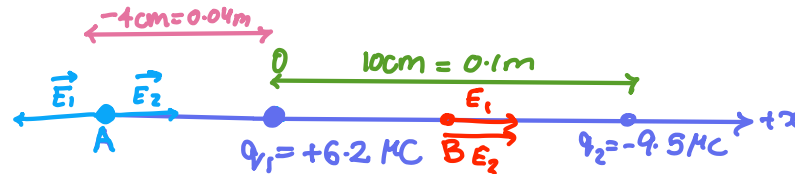
Exercise

Two point-charges lie along the x-axis. A charge of $+6.2 \mu\text{C}$ is at the origin and a charge of $-9.5 \mu\text{C}$ is placed at a distance $x = 10 \text{ cm}$.

What is the net electric field at:

a) $x = -4.0 \text{ cm}$ **A**

b) $x = +4.0 \text{ cm}$ **B**



$$\textcircled{a} \quad E_1 = \frac{k|q_1|}{r^2} = \frac{9 \times 10^9 |6.2 \times 10^{-6}|}{(0.04)^2} = 3.49 \times 10^7 \frac{\text{N}}{\text{C}} (-\hat{x}) \text{ or } 34875000$$

$$E_2 = \frac{k|q_2|}{r^2} = \frac{9 \times 10^9 |9.5 \times 10^{-6}|}{(0.14)^2} = 4.36 \times 10^6 \frac{\text{N}}{\text{C}} (+\hat{x}) \approx 4362244.8$$

$$E_{\text{net}} = -3.49 \times 10^7 + 4.36 \times 10^6 = -3.05 \times 10^7 \frac{\text{N}}{\text{C}} (-\hat{x})$$

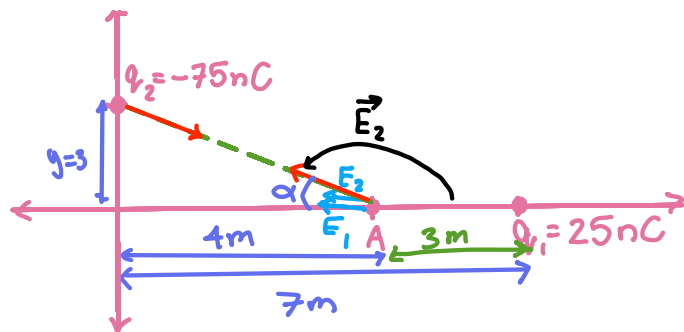
$$\textcircled{b} \quad E_1 = \frac{k q_1}{r^2} = \frac{9 \times 10^9 |6.2 \times 10^{-6}|}{(0.04)^2} = 3.49 \times 10^7 \frac{\text{N}}{\text{C}} (-\hat{x})$$

$$E_2 = \frac{9 \times 10^9 |9.5 \times 10^{-6}|}{(0.06)^2} = 2.38 \times 10^7 \frac{\text{N}}{\text{C}}$$

$$E_{\text{net}} = 3.49 \times 10^7 + 2.38 \times 10^7 = 5.87 \times 10^7 \frac{\text{N}}{\text{C}} (+\hat{x})$$

Exercise

A charge of 25 nC is placed on the x-axis at $x = 7$ m and a charge of -75 nC is placed on the y-axis at $y = 3$ m. What is the magnitude and direction of the net electric field produced at the point $x = 4$ m?



$$n = 10^{-9}$$

$$E_1 = \frac{k|q|}{r^2} = \frac{9 \times 10^9 |25 \times 10^{-9}|}{(3)^2} = 25 \frac{\text{N}}{\text{C}} (-\hat{x})$$

$$E_2 = \frac{k|q|}{r^2} = \frac{9 \times 10^9 |75 \times 10^{-9}|}{(5)^2} = 27 \frac{\text{N}}{\text{C}}$$

$$\alpha = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{3}{4}\right) = 37^\circ$$

Pythagorean theorem

$$\vec{E}_2 (27 \frac{\text{N}}{\text{C}}, 180 - 37) = (27 \frac{\text{N}}{\text{C}}, 143^\circ)$$

$$E_{2x} = E_2 \cos \alpha = 27 \cos(143) = -21.5 \frac{\text{N}}{\text{C}} (-\hat{x})$$

$$E_{2y} = E_2 \sin \alpha = 27 \sin(143) = 16.25 \frac{\text{N}}{\text{C}} (+\hat{y})$$

$$E_{\text{net}x} = -21.5 - 25 = -46.5 \frac{\text{N}}{\text{C}}$$

$$E_{\text{net}y} = 16.25 \frac{\text{N}}{\text{C}} (+\hat{y})$$

$$E_{\text{net}} = \sqrt{(E_{\text{net}x})^2 + (E_{\text{net}y})^2} = 49.27 \frac{\text{N}}{\text{C}} \quad \theta = \tan^{-1}\left(\frac{16.25}{-46.5}\right) + 180 = 160.83^\circ$$

3 cases:

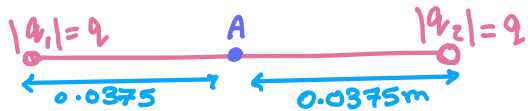
- ① 1 component
- ② Perpendicular
- ③ Forces neither x nor y

put negative in $\tan^{-1}\left(\frac{y}{x}\right)$
then add 180

Exercise (HW)

Two point-charges of equal magnitudes are 7.5 cm apart. At the midpoint of the line connecting them, their combined electric field has a magnitude of 45 N/C. Find the magnitude of the charges.

$$|q_1| = |q_2| = q$$



$$7.5 \text{ cm} = 0.075 \text{ m}$$

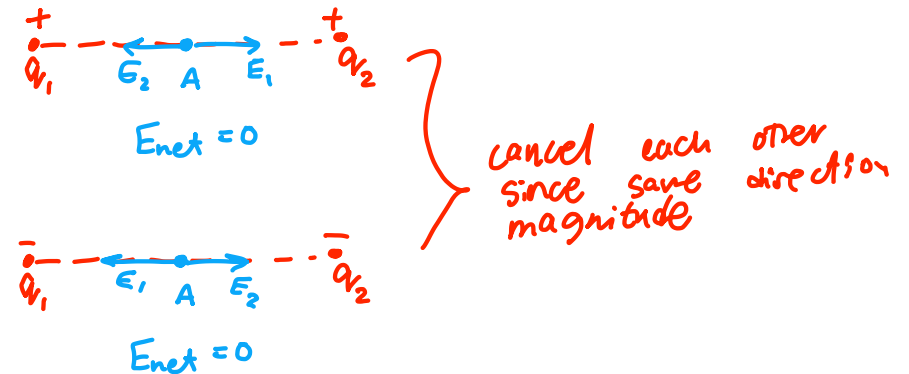
$$\begin{aligned} E_{\text{net}A} &= 45 \frac{\text{N}}{\text{C}} = E_1 + E_2 \\ &= k \frac{|q_1|}{r^2} + k \frac{|q_2|}{r^2} \\ &= \frac{kq}{r^2} + \frac{kq}{r^2} \\ &= \frac{2kq}{r^2} \end{aligned}$$

$$45 = \frac{2(9 \times 10^9)q}{(0.0375)^2}$$

* solve for q *

$$q = 3.5 \times 10^{-12} \text{ C}$$

pico



Solution

The charges must have opposite signs, or their electric fields would cancel out at the point midway between them.

- $E_{\text{tot}} = E_{q1} + E_{q2} = 2 E_q = 45$ (but remember that electric fields are vectors)
- $22.5 = k |q| / (3.75 \times 10^{-2})^2$

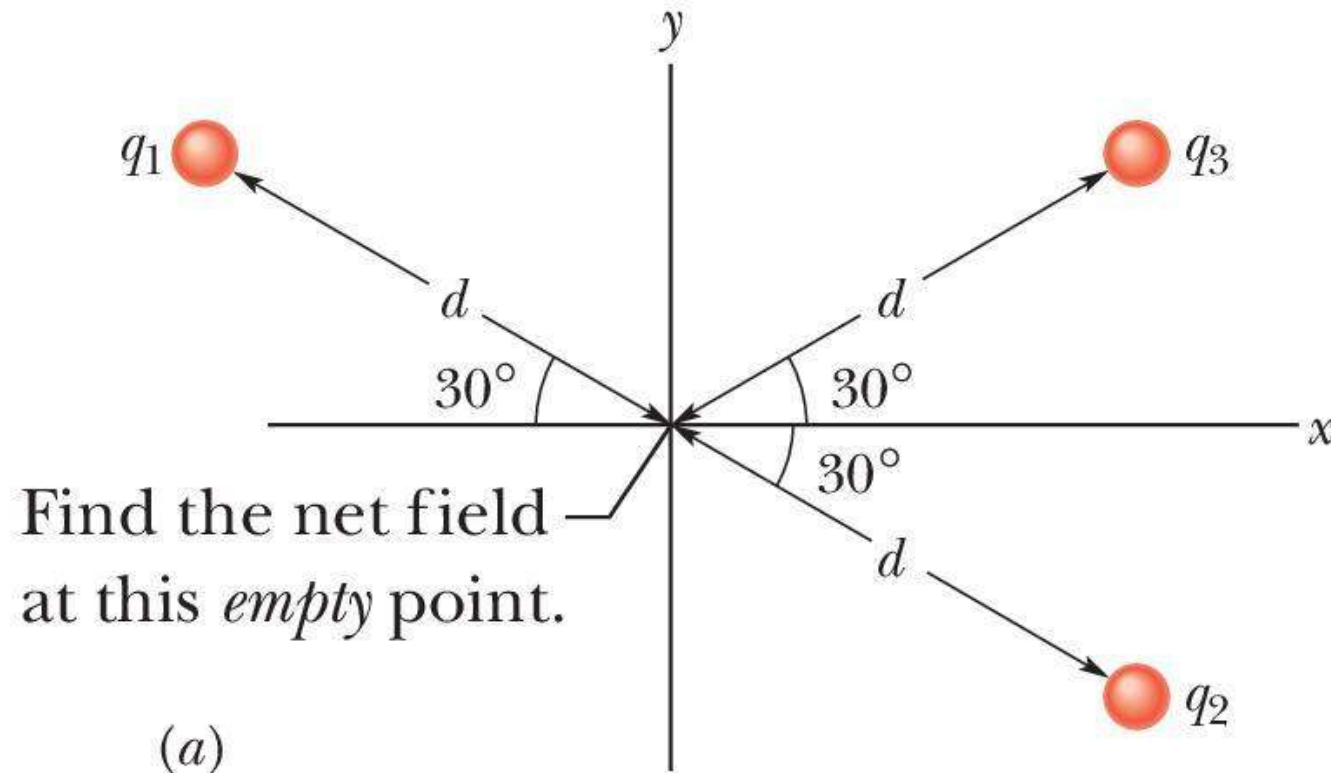
Then,

$$|q| = 3.52 \times 10^{-12} \text{ C}$$

Extra Home Practice

Find the net electric field at the origin.

Given $q_1 = +2Q$, $q_2 = -2Q$, and $q_3 = -4Q$



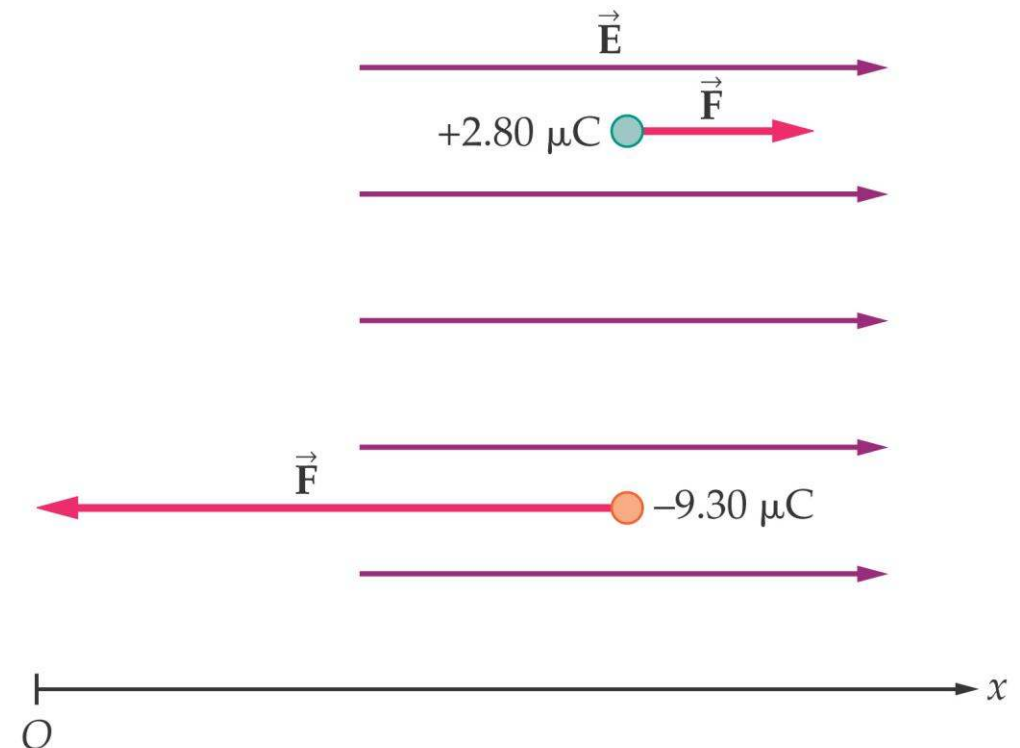
Section 22.6 Force on a Point Charge in an Electric Field

- If a particle with charge q is placed in an external electric field $\vec{E}$, an electrostatic force $\vec{F}$ acts on the particle:

$$\vec{F} = q\vec{E}$$

same *+ direction*
opposite *- direction*

- The force acting on the charged particle has the **same direction** of $\vec{E}$ if the charge of the particle q is **positive**, and has the **opposite direction** if the charge q is **negative**



Exercise: A $+5.0 \mu\text{C}$ charge experiences a 0.44 N force in the positive y -direction. If this charge is replaced with a $-2.7 \mu\text{C}$ charge, what force will it experience?

$$\begin{aligned}
 F &= Eq \\
 q_1 &= +5 \times 10^{-6} \text{ C} \\
 F &= 0.44 \text{ N} \\
 \frac{0.44}{5 \times 10^{-6}} &= \frac{E (5 \times 10^{-6})}{5 \times 10^{-6}} \\
 E &= 88000 \frac{\text{N}}{\text{C}} (+y) \\
 F &= 88000 \times |-2.7 \times 10^{-6}| \\
 F &= 0.2376 \text{ N } (-y)
 \end{aligned}$$

Small space, 1 line (not much 'space')

part a $\rightarrow$ same as question above

part b q_3 added to question after finding E_{net} find

net Force $\begin{matrix} +: F \& E \text{ same direction} \\ -: F \& E \text{ opposite} \end{matrix}$

$$F = |q_3| E_A$$

Solution:

- Find the electric field:

$$E = F/|q| = 0.44 / (5 \times 10^{-6}) = 8.8 \times 10^4 (+\hat{y}) \text{ N/C}$$

- Now, the force on the second charge is:

$$F = |q| E = (2.7 \times 10^{-6}) \times (8.8 \times 10^4) = 23.8 \times 10^{-2} \Rightarrow 0.24 (-\hat{y}) \text{ N/C}$$

** The **field is exactly the same** at that point **regardless of**

1) **The sign** or

2) Magnitude of **the charge** that is located there.

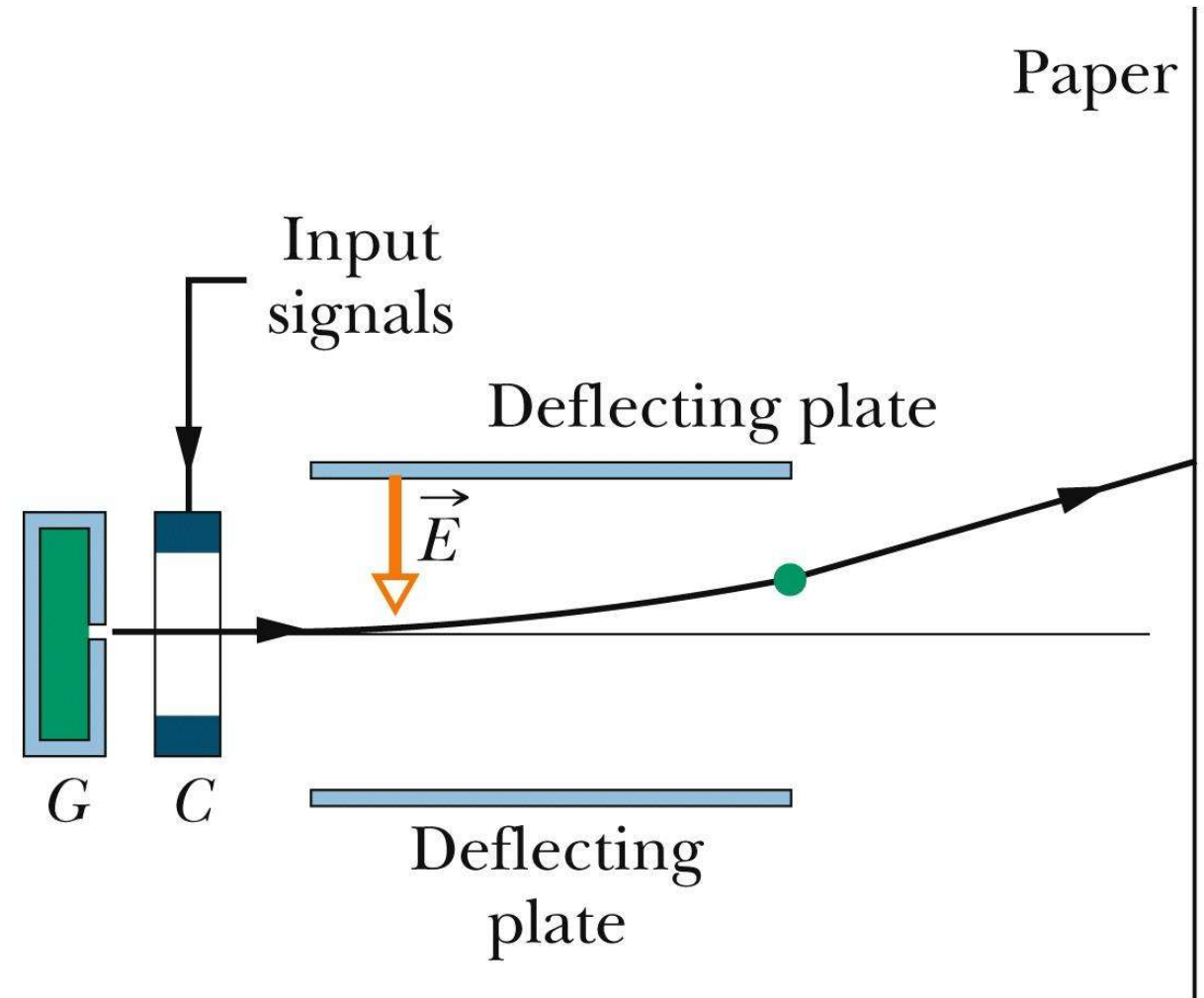
Ink-jet Printer Application

- Drops shot from generator G receive a charge in charging unit C . An input signal from a computer controls the charge and thus the effect of field $\vec{E}$ on where the drop lands on the paper.

• fingerprints (powder)

• car painting

↳ maintain consistency



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